

About Me

I am a Ph.D. student major in Computer Science in [BDSC lab](#) at [University of Illinois Chicago \(UIC\)](#), where I am fortunate to be supervised by [Prof. Philip S. Yu](#).

I completed my master's degree in Mechanical Engineering from [Dalian University of Technology \(DUT\)](#), and obtained my bachelor's degree from [China University of Petroleum \(UPC\)](#). I am grateful to Prof. Wenhao Luo for his mentorship and guidance during my earlier doctoral research in robotics and safe autonomous systems.

My research interests primarily lie in Robotics, Reinforcement Learning, Computer Vision and Control Theory. My current research focuses on addressing fundamental challenges in grounding large language models (LLMs) in robotics, with the goal of enabling intelligent agents to effectively perceive, reason about, and interact with the physical world.

News

- **[July 2026]** Our paper about adaptive uncertainty quantification is accepted by [CDC 2026](#).
- **[April 2026]** Our paper about safe model-based RL is accepted by [IROS 2026](#).
- **[April 2026]** Our paper about Time-aggregated Connectivity is accepted by [RSS 2026](#).
- **[April 2026]** Our paper about Adaptive PT-CBF is accepted by [IFAC 2026](#).
- **[Jan. 2026]** Our paper about geometry-aware CBF is accepted by [ICRA 2026](#).
- **[Jun. 2025]** I will present our Multi-robot Deadlock Resolution paper in [MMLS 2025](#).
- **[Jan. 2025]** Our paper about Multi-robot Deadlock Resolution is accepted to [ICRA' 25](#).
- **[Jan. 2025]** Our paper about Sample-efficient Safe Reinforcement Learning is accepted to [ICRA' 25](#).
- **[Nov. 2024]** I will attend the [Join the 5-Day Gen AI Intensive Course with Google](#).
- **[Oct. 2024]** Our project [Turbo Trash](#) obtained the 1st place in HACKTHON CLT 2024.
- **[Sep. 2024]** I will present our work about autonomous driving on ITSC 2024.

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- **[Jul. 2024]** Our paper about Courteous Driving using CBF-inspired Risk is accepted to [ITSC' 24](#).
- **[Jul. 2024]** Our paper about Individualizable Risk Assessment is accepted to [ITSC' 24](#).
- **[June. 2024]** Our paper about Data-driven Connectivity Maintenance is accepted to [IROS' 24](#).
- **[June. 2024]** Our paper about Vehicle Longitudinal Motion Prediction is accepted to [MECC' 24](#).
- **[May. 2024]** Our paper about Line-of-Sight Connectivity Maintenance is accepted to [RSS'24](#).
- **[Jan. 2024]** Our paper about Conformal Prediction is accepted to [ACC'24](#).
- **[Oct. 2023]** I will present our work at [MRS'23](#).
- **[Oct. 2023]** Our paper about multi-agent deadlock resolution is accepted to [MRS'23](#).
- **[Oct. 2023]** Our paper about connectivity maintenance is accepted to [MRS'23](#).
- **[Sep. 2023]** Our paper about connectivity maintenance is accepted to [IROS'23 Workshop](#).
- **[Mar. 2023]** Our paper about Image-Based Visual Servoing is accepted to [IFAC'23](#).



Highlights



demo

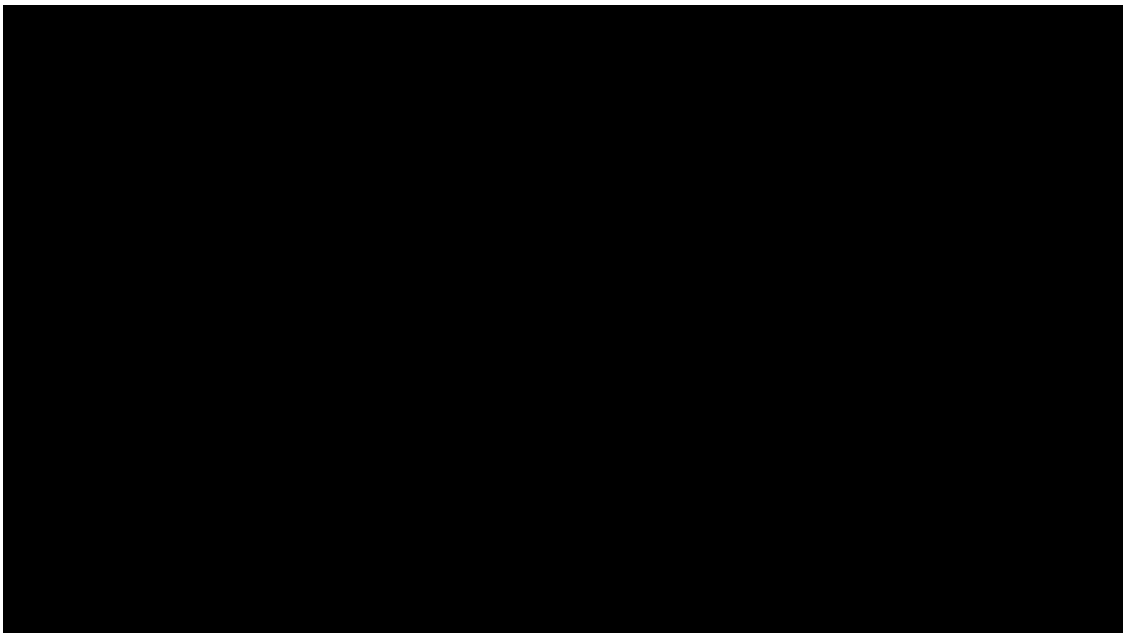
Adaptive Deadlock Avoidance for Decentralized Multi-Agent Systems via CBF-inspired Risk Measurement

Yanze Zhang, Yiwei Lyu, Siwon Jo, Yupeng Yang, Wenhao Luo

IEEE International Conference on Robotics and Automation (ICRA), 2025

Paper

- We present a novel decentralized framework that ensures both efficient task execution and deadlock resolution for multi-agent systems, which enables co-optimization between the task-related controller and deadlock resolution controller, yielding smoother robots' motion with improved overall task execution efficiency



****Courteous MPC for Autonomous Driving with CBF-inspired Risk Assessment****

Yanze Zhang, Yiwei Lyu, Sude E. Demir, Xingyu Zhou, Yupeng Yang, Junming Wang, Wenhao Luo

IEEE International Conference on Intelligent Transportation Systems (ITSC), 2024

Paper

- we develop an extension of the CBF-inspired risk evaluation framework that takes into account both noisy observed positions and motions, which are then integrated with MPC to ensure smooth and robust decision-making for AVs.

All Publications

- *Adaptive Deadlock Avoidance for Decentralized Multi-Agent Systems via CBF-inspired Risk Measurement.*

Yanze Zhang, Yiwei Lyu, Siwon Jo, Yupeng Yang, and Wenhao Luo

Accepted to the 2025 IEEE International Conference on Robotics and Automation (**ICRA' 25**), 2025

[\[PDF\]](#)

- *Computation and Sample Efficient Safe Reinforcement Learning Using Adaptive Conformal Prediction.*

Hao Zhou, **Yanze Zhang**, and Wenhao Luo

Accepted to the 2025 IEEE International Conference on Robotics and Automation (**ICRA' 25**), 2025

[\[PDF\]](#)

- *Courteous MPC for Autonomous Driving with CBF-inspired Risk Assessment.*

Yanze Zhang, Yiwei Lyu, Sude Ela Demir, Xingyu Zhou, Yupeng Yang, Junmin Wang and Wenhao Luo

Accepted to the 27th IEEE International Conference on Intelligent Transportation Systems (**ITSC' 24**), 2024

[\[PDF\]](#)[\[Demo-NGSIM\]](#)[\[Demo-IDM\]](#)

- *Individualizable Risk Assessment Map for Planning Vehicle Behaviors Respecting Perceived Safety.*

Ji Hwan Park, **Yanze Zhang**, Wenhao Luo, and Junmin Wang

Accepted to the 27th IEEE International Conference on Intelligent Transportation Systems (**ITSC'**

24), 2024

[\[PDF\]](#)

- *Integrating Online Learning and Connectivity Maintenance for Communication-Aware Multi-Robot Coordination.*

Yupeng Yang, Yiwei Lyu, **Yanze Zhang**, Ian Gao, and Wenhao Luo

Accepted to the 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS' 24**), 2024

[\[PDF\]](#)[\[Project\]](#)[\[Website\]](#)

- *A Transitional Intelligent Driver Model Enabling Vehicle Longitudinal Motion Prediction in Lane-Change Maneuvers.*

Sude E. Demir, Xingyu Zhou, **Yanze Zhang**, Wenhao Luo, and Junmin Wang

Accepted to the 4th Modeling, Estimation, and Control Conference (**MECC' 24**), 2024

[\[PDF\]](#)

- *Decentralized Multi-Robot Line-of-Sight Connectivity Maintenance under Uncertainty.*

Yupeng Yang, Yiwei Lyu, **Yanze Zhang**, Sha Yi, and Wenhao Luo

Accepted to the 2024 Robotics: Science and Systems (**RSS'24**), 2024

[\[PDF\]](#)[\[Video\]](#)

- *Safety-Critical Control with Uncertainty Quantification using Adaptive Conformal Prediction.*

Hao Zhou, **Yanze Zhang**, and Wenhao Luo

[\[PDF\]](#)

- *Occlusion-Free Image Based Visual Servoing Using Probabilistic Control Barrier Certificates.*

Yanze Zhang, Yupeng Yang, Wenhao Luo

Accepted to appear in the 22th World Congress of International Federation of Automatic Control (**IFAC'23**), 2023

[\[PDF\]](#)[\[Slides\]](#)



Honors and Awards

- 2023.06 Conference Travel Award from the Graduate School of UNC Charlotte.
- 2017, 2019 [RoboMaster](#) International First Price.



Education

- 2022.08 - Now, Doctor of Philosophy, Computer Science, University of North Carolina at Charlotte, The United States
- 2019.09 - 2022.06, Master of Engineering, Mechanical Design and Theory, Dalian University of Technology, China
- 2015.09 - 2019.06, Bachelor of Engineering, Mechanical Engineering, China University of Petroleum, China



Services

External Reviewer

- IEEE Conference on Decision and Control (CDC' 26)
- The IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS' 26)
- The 2026 American Control Conference (ACC' 26)
- IEEE Transactions on Automation Science and Engineering (IEEE T-ASE)
- IEEE International Symposium on Multi-Robot & Multi-Agent Systems (IEEE MRS' 25)
- The IEEE/ASME Transactions on Mechatronics (TMeCh)
- The Conference on Robot Learning (CoRL' 25, CoRL' 26)
- IEEE Robotics and Automation Letters (R-AL)
- 12th IFAC Symposium on Intelligent Autonomous Vehicles (IAV' 25)

- IEEE International Conference on Robotics and Automation (ICRA' 25, ICRA' 26)
- International Federation of Automatic Control (IFAC'23, IFAC'26)

Mentoring Activities

- [Ian Gao](#), Bachelor Research Project. Now at UNC Chaphill as an undergraduate.
- John Modl, [REU Program](#).
- [Matthew De La Rosa](#), [REU Program](#).
- [Andrew Chen](#), [REU Program](#). Now at [CMU](#) as a Master student.
- Zachary Palko, [REU Program](#).
- [Siwon Jo](#), Bachelor Research Project. Now at University of Pennsylvania as a Master student.
- [Mukul Mohan](#), Bachelor Research Project.

(last updated: Jun. 2026)